Alur Aplikasi - Sistem Tracking Status Dokumen Notaris

1. Overview Alur Aplikasi

Dokumen ini menjelaskan alur aplikasi secara naratif untuk tiga use case utama:

1. Klien cek status dokumen
2. Sistem ambil data dan validasi
3. Notaris/staff update status

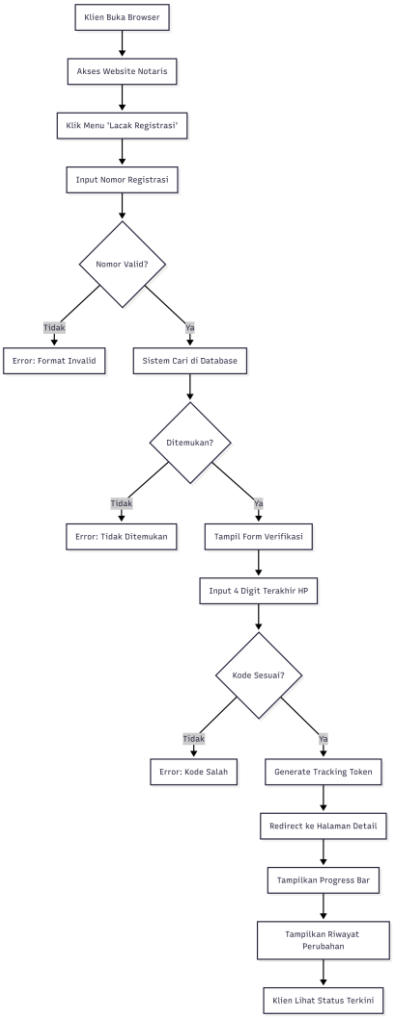
2. Alur 1: Klien Cek Status Dokumen

2.1 Narasi Lengkap

Aktor: Klien (pengguna layanan notaris)

Tujuan: Klien ingin mengetahui status progress dokumen mereka secara mandiri tanpa harus menghubungi kantor notaris.

Precondition: Klien memiliki nomor registrasi yang diberikan saat pendaftaran dokumen.



2.2 Step-by-Step Detail

Step 1: Akses Halaman Tracking

- User Action: Klien membuka browser dan mengakses website Kantor Notaris Sri Anah
- System Response: Homepage ditampilkan dengan menu navigasi
- Navigation: Klien klik menu "Lacak Registrasi" di navigation bar
- URL: /index.php?gate=lacak

Step 2: Input Nomor Registrasi

- User Action: Klien mengetik nomor registrasi pada form search
- Example Input: NP-20260326-1234
- System Action: Validasi format (harus NP-YYYYMMDD-XXXX)

Step 3: Validasi Nomor Registrasi

System Process:

1. Sanitasi input (hapus karakter berbahaya)
2. Query database: SELECT * FROM registrasi WHERE nomor_registrasi = 'NP-20260326-1234'

3. Check hasil query

Kemungkinan Hasil:

- Tidak Valid Format: Error "Format nomor registrasi tidak valid"
- Tidak Ditemukan: Error "Nomor registrasi tidak ditemukan"
- Ditemukan: Lanjut ke verifikasi

Step 4: Verifikasi 4 Digit HP

- System Response: Form verifikasi ditampilkan
- User Action: Klien input 4 digit terakhir nomor HP mereka
- Example: Jika HP klien 0812-3456-7890, input: 7890

Security Measure:

- Rate limiting: Maksimal 5 percobaan per menit
- Logging failed attempts untuk deteksi brute force

Step 5: Validasi Kode Verifikasi

System Process:

1. Ambil data klien dari registrasi

2. Extract 4 digit terakhir dari nomor HP (remove non-numeric first)
3. Bandingkan dengan input klien
4. Jika match → generate token
5. Jika tidak match → error

Contoh:

- HP di Database: 081234567890
- Clean HP: 081234567890
- Last 4: 7890
- Input Klien: 7890
- Result: MATCH ✅

Step 6: Generate Tracking Token

System Process:

1. Create payload: {id: registrasi_id, code: verification_code, exp: time + 24h}
2. Encode base64
3. Generate HMAC-SHA256 signature
4. Format: base64Payload.signature
5. Simpan token di database (registrasi.tracking_token)

Token Example:

`eyJpZCI6MTIzLCJjb2RlIjoiYTFiMmMzZDQiLCJleHAiOjE3MTY5ODc2NTR9.abc123def456...`

Step 7: Tampilkan Progress Tracking

- System Response: Redirect ke halaman detail dengan token
- URL: /index.php?gate=detail&token=eyJpZCI6...

Page Content:

- Nomor registrasi
- Nama klien (disamarkan sebagian)
- Layanan
- Status saat ini
- Progress bar 15 status
- Estimasi waktu
- Riwayat perubahan (timeline)

Step 8: Lihat Riwayat (Process Log)

- System Query: SELECT * FROM registrasi_history WHERE registrasi_id = ? ORDER BY created_at DESC

Display Format:

26 Mar 2026, 14:30

Status: Pembayaran Admin → Validasi Sertifikat

Catatan: Berkas lengkap, lanjut validasi

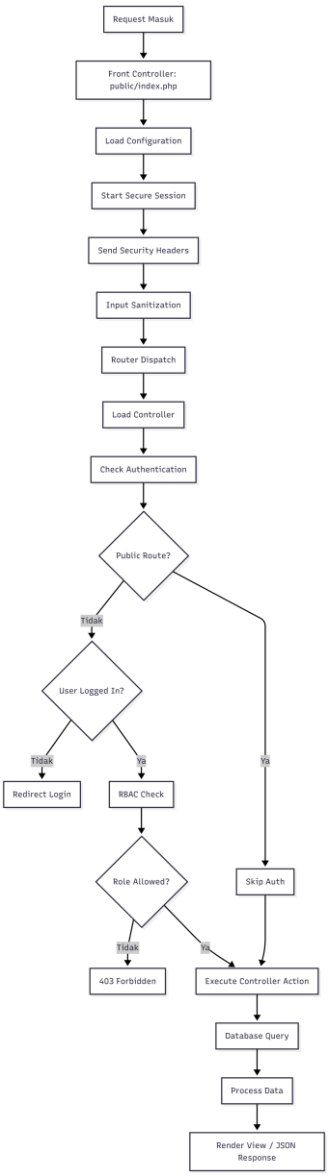
Oleh: Admin

3. Alur 2: Sistem Ambil Data dan Validasi

3.1 Narasi Lengkap

Aktor: Sistem (Backend)

Tujuan: Mengambil data registrasi dari database dan melakukan validasi keamanan.



3.2 Request Lifecycle Detail

Phase 1: Front Controller (public/index.php)

```
// 1. Define BASE_PATH
define('BASE_PATH', dirname(__DIR__));

// 2. Register Autoloader
App\Core\Autoloader::register();
App\Core\Autoloader::addNamespace('App\\', BASE_PATH . '/app/');
App\Core\Autoloader::addNamespace('Modules\\', BASE_PATH . '/modules/');

// 3. Load Configuration
require_once BASE_PATH . '/config/app.php';

// 4. Load Utils
require_once BASE_PATH . '/app/Core/Utils/helpers.php';
require_once BASE_PATH . '/app/Core/Utils/security.php';

// 5. Secure Session
App\Security\Auth::startSecureSession();

// 6. Security Headers
sendSecurityHeaders();
header('Cache-Control: no-cache, no-store, must-revalidate');

// 7. Input Sanitization
App\Security\InputSanitizer::sanitizeGlobal();

// 8. Load Routes
require_once BASE_PATH . '/config/routes.php';

// 9. Dispatch Router
App\Core\Router::dispatch();
```

Phase 2: Router Dispatch

```
// Router::dispatch()
$gate = $_GET['gate'] ?? 'home';
$method = $_SERVER['REQUEST_METHOD'];

// Lookup route
$route = self::$routes[$gate][$method] ?? null;

if ($route) {
    // Check auth requirement
    if ($route['auth'] && !Auth::check()) {
        redirect('/index.php?gate=login');
        exit;
    }

    // Check RBAC
    if (isset($route['role'])) {
        RBAC::enforce($route['role']);
    }

    // Execute controller
    [$controllerClass, $action] = $route['handler'];
    $controller = new $controllerClass();
    $controller->$action();
}
```

Phase 3: Database Query

```
// Example: Tracking search
public function findByNomorRegistrasi($nomor) {
    return Database::selectOne(
        "SELECT p.id, p.klien_id, p.nomor_registrasi, p.status,
                k.nama AS klien_nama, k.hp AS klien_hp,
                l.nama_layanan
         FROM registrasi p
         LEFT JOIN klien k ON p.klien_id = k.id
         LEFT JOIN layanan l ON p.layanan_id = l.id
```

```
         WHERE p.nomor_registrasi = :nomor
         LIMIT 1",
        ['nomor' => $nomor]
    );
}
```

Phase 4: Token Validation

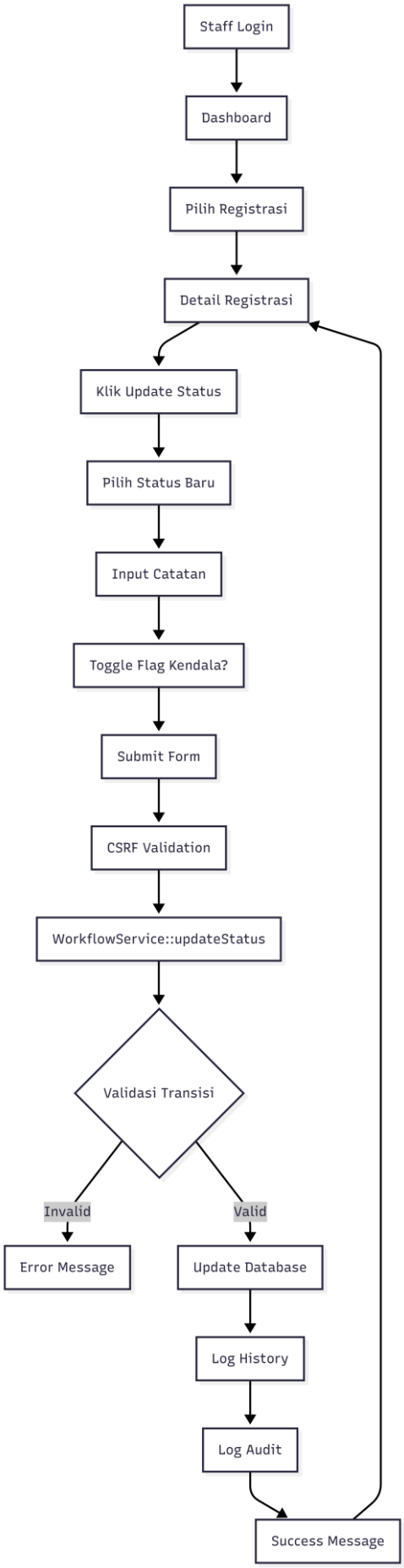
```
// verifyTrackingToken()
function verifyTrackingToken($token) {
    // 1. Parse token
    $parts = explode('.', $token);
    if (count($parts) !== 2) return false;

    // 2. Decode payload
    $payload = base64_decode($parts[0]);
    $data = json_decode($payload, true);
    if (!$data) return false;

    // 3. Verify signature
    $expectedSig = hash_hmac('sha256', $parts[0], SECRET_KEY);
    if (!hash_equals($expectedSig, $parts[1])) return false;

    // 4. Check expiration
    if (isset($data['exp']) && $data['exp'] < time()) return false;

    return $data;
}
```

4. Alur 3: Notaris/Staff Update Status

4.1 Narasi Lengkap

Aktor: Staff Admin / Notaris

Tujuan: Update status progress dokumen dengan validasi workflow.


Staff Login
Dashboard
Pilih Registrasi
Detail Registrasi
Klik Update Status
Pilih Status Baru
Input Catatan
Toggle Flag Kendala?
Submit Form
CSRF Validation
WorkflowService::updateStatus
Validasi Transisi
Invalid
Valid
Error Message
Update Database
Log History
Log Audit
Success Message

4.2 Step-by-Step Detail

Step 1: Authentication

User Action: Staff login dengan username dan password

System Process:

1. Query user by username
2. password_verify() dengan bcrypt hash
3. Start secure session dengan fingerprinting
4. Log audit: 'login'
5. Redirect ke dashboard

Session Security:

```
// Session fingerprinting

$fingerprint = hash('sha256', $_SERVER['HTTP_USER_AGENT'] .
$_SERVER['REMOTE_ADDR']);

$_SESSION['user_fingerprint'] = $fingerprint;


// Validate on each request

if ($_SESSION['user_fingerprint'] !== $currentFingerprint) {

    session_destroy(); // Hijacking detected!

}
```

Step 2: Akses Detail Registrasi

- Navigation: Dashboard → Registrasi → Klik nomor registrasi
- URL: /index.php?gate=registrasi_detail&id=123
- Controller: Dashboard\Controller::showRegistrasi()

Database Query:

```
  SELECT registrasi.*, klien.*, layanan.*

  FROM registrasi

  LEFT JOIN klien ON registrasi.klien_id = klien.id

  LEFT JOIN layanan ON registrasi.layanan_id = layanan.id

  WHERE registrasi.id = 123
```

Step 3: Update Status Form

```
<!-- Simplified form structure -->

<form method="POST" action="/index.php?gate=update_status">

    <input type="hidden" name="csrf_token" value="abc123...">

    <input type="hidden" name="registrasi_id" value="123">
```

```
    <select name="status">
        <option value="draft">Draft</option>
        <option value="pembayaran_admin">Pembayaran Admin</option>
        <!-- ... other statuses ... -->
    </select>

    <textarea name="catatan"></textarea>

    <label>
        <input type="checkbox" name="flag_kendala">
        Flag Kendala
    </label>

    <button type="submit">Update Status</button>
</form>
```

Step 4: Workflow Validation

```
// WorkflowService::updateStatus()

// 1. Load registrasi
$registrasi = $this->registrasiModel->findById($registrasiId);

// 2. Check final status
if (in_array($registrasi['status'], ['selesai', 'ditutup', 'batal'])) {
    return ['success' => false, 'message' => 'Status final tidak bisa diubah'];
}

// 3. Check locked
if ($registrasi['is_locked']) {
    return ['success' => false, 'message' => 'Registrasi sedang dikunci'];
}

// 4. Get status order
$currentOrder = STATUS_ORDER[$registrasi['status']];
$newOrder = STATUS_ORDER[$newStatus];
```

```
// 5. Check backward transition
if ($newOrder < $currentOrder && $newStatus !== 'batal') {
    if ($registrasi['status'] !== 'perbaikan') {
        return ['success' => false, 'message' => 'Status tidak bisa mundur'];
    }
}

// 6. Check cancellation
if ($newStatus === 'batal') {
    if (!in_array($registrasi['status'], CANCELLABLE_STATUSES)) {
        return ['success' => false, 'message' => 'Tidak bisa batal setelah pajak'];
    }
}

// 7. Update database
$this->registrasiModel->updateStatus($registrasiId, $newStatus, $keterangan,
$catatan);

// 8. Handle flag kendala
if ($flagKendala !== null) {
    // Toggle kendala logic
}

// 9. Save history
$this->registrasiHistoryModel->create([
    'registrasi_id' => $registrasiId,
    'status_old' => $registrasi['status'],
    'status_new' => $newStatus,
    'catatan' => $catatan,
    'user_id' => $userId,
    'user_name' => $userName,
    'user_role' => $role,
    'ip_address' => $_SERVER['REMOTE_ADDR'],
]);
```

```
// 10. Audit log
AuditLog::create([
    'user_id' => $userId,
    'role' => $role,
    'action' => 'update',
    'old_value' => json_encode(['status' => $registrasi['status']]),
    'new_value' => json_encode(['status' => $newStatus]),
]);
```

Step 5: Status Transition Examples

Valid Transitions:

✅ draft → pembayaran_admin (forward)

✅ pembayaran_admin → validasi_sertifikat (forward)

✅ draft → batal (cancellable)

✅ perbaikan → pembayaran_pajak (loop back allowed)

Invalid Transitions:

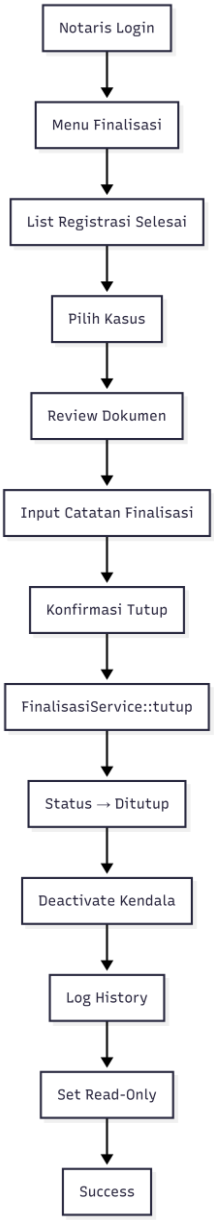
❌ validasi_sertifikat → draft (backward not allowed)

❌ pembayaran_pajak → batal (past cancellation limit)

❌ selesai → pendaftaran (final status read-only)

5. Alur 4: Notaris Finalisasi Kasus

5.1 Narasi Lengkap



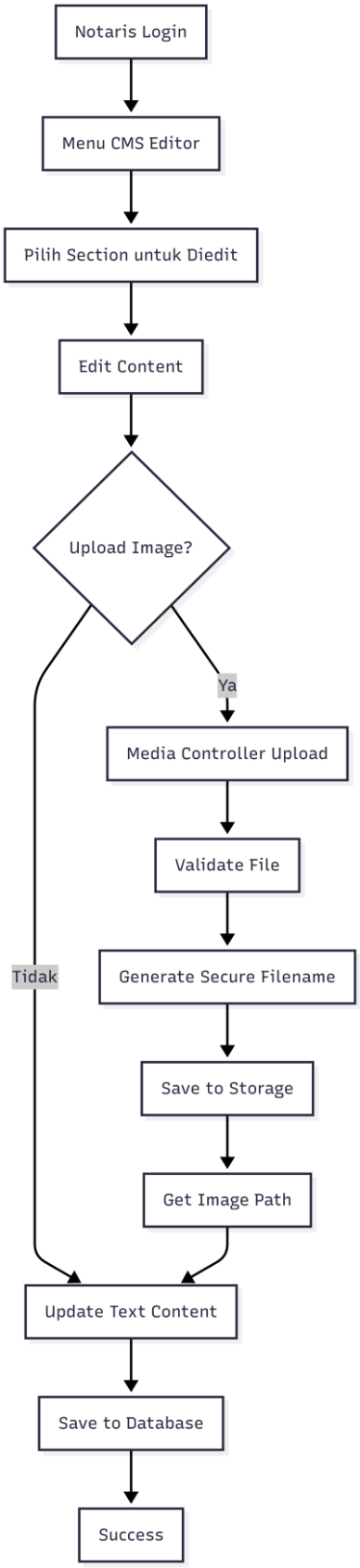
5.2 Detail Process

```
// FinalisasiService::tutupRegistrasi()


public function tutupRegistrasi($registrasiId, $userId, $role, $notes = null) {

    // 1. Load registrasi

    $registrasi = $this->registrasiModel->findById($registrasiId);


    // 2. Check status must be 'selesai'

    if ($registrasi['status'] !== 'selesai') {

        return ['success' => false, 'message' => 'Status harus selesai untuk
finalisasi'];

    }
```

```
    // 3. Update status to 'ditutup'
    $this->registrasiModel->update($registrasiId, [
        'status' => 'ditutup',
        'is_finalized' => 1,
        'finalized_at' => date('Y-m-d H:i:s'),
        'finalized_by' => $userId,
        'finalization_notes' => $notes,
    ]);

    // 4. Deactivate all kendala flags
    $this->kendalaModel->deactivateAll($registrasiId);

    // 5. Save history
    $this->registrasiHistoryModel->create([
        'registrasi_id' => $registrasiId,
        'status_old' => 'selesai',
        'status_new' => 'ditutup',
        'catatan' => $notes,
        'user_id' => $userId,
        'user_name' => $userName,
        'user_role' => $role,
    ]);

    // 6. Audit log
    AuditLog::create([
        'user_id' => $userId,
        'role' => $role,
        'action' => 'finalize',
        'old_value' => json_encode(['status' => 'selesai']),
        'new_value' => json_encode(['status' => 'ditutup']),
    ]);

    return ['success' => true, 'message' => 'Kasus berhasil ditutup'];
}
```

6. Alur 5: CMS Management

6.1 Narasi



6.2 Upload Flow

```
// Media\Controller::upload()

public function upload() {
    // 1. Check auth
    RBAC::enforce('notaris');

    // 2. Validate file
    if (!isset($_FILES['image'])) {
        return json(['success' => false, 'message' => 'No file']);
    }

    $file = $_FILES['image'];

    // 3. Check size (max 5MB)
    if ($file['size'] > MAX_UPLOAD_SIZE) {
        return json(['success' => false, 'message' => 'File too large']);
    }

    // 4. Check extension
    $ext = strtolower(pathinfo($file['name'], PATHINFO_EXTENSION));
    if (!in_array($ext, ['jpg', 'jpeg', 'png', 'pdf'])) {
        return json(['success' => false, 'message' => 'Invalid file type']);
    }

    // 5. Generate secure filename
    $secureName = 'img_' . bin2hex(random_bytes(16)) . '.' . $ext;

    // 6. Move file
    $destination = PUBLIC_PATH . '/assets/images/' . $secureName;
    move_uploaded_file($file['tmp_name'], $destination);

    // 7. Return path
    return json([
        'success' => true,
```

```
        'path' => '/assets/images/' . $secureName,
    ]);
}
```

7. Kesimpulan

Alur aplikasi yang telah diuraikan mencakup:

1. Tracking Klien - Full flow dari search hingga viewing progress
2. System Processing - Request lifecycle dari front controller hingga database
3. Update Status - Workflow validation dengan business rules
4. Finalisasi - Tutup kasus dengan auto-deactivate kendala
5. CMS Management - Content editing dan image upload

Setiap alur mengikuti prinsip:

- Security First - CSRF, RBAC, input sanitization
- Business Rules - Workflow validation, status order
- Audit Trail - Logging semua aksi penting
- User Experience - Clear error messages, success feedback